

MACHINEPULSE · REQUIREMENTS

MachinePulse requirements

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DOCUMENT CONTROL

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Project TRL	TRL 2 Technology concept formulated		
Repository	github.com/BoujeeEnjinia1701/machinepulse		
Source	docs/03-requirements.md @ e0901e9		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with concept status
0.1	2026-09-25	Amish Chadha	Initial scaffold

MachinePulse requirements

These are first-pass requirements for the concept. Targets are proposals for review, awaiting Amish. The status column gives the first-order view from the design precis (MPL-PRC-001); every "met" is an estimate to be checked by calculation at TRL 3. Three requirements are not met by the concept as it stands (R4, R10 and R12), and two are unverified with a known risk (R6 and R11).

Table 1. Requirements and concept status.

ID	Requirement	Target	Concept status (estimate)	Verification (TRL 3 or later)
R1	Detect run state	Off, idle and running classified from current, with 1 s time resolution	Met: CT RMS every 1 s against two thresholds	Calculation; later bench test on a motor
R2	Log run time	Run hours per machine, error within 1 % over a week	Met: 1 s resolution gives far less than 1 % error on runs over 2 min	Calculation
R3	Measure load current	Within 5 % of reading from 10 % to 100 % of CT range, after a two-point calibration	Met on paper; ESP32 ADC nonlinearity is the main risk	Error budget; later comparison with a clamp meter
R4	Estimate energy per shift	Within 10 % of a reference meter	Not met. One CT with assumed voltage and power factor gives about 20 %; needs voltage sensing or three CTs	Error budget
R5	Measure vibration	Velocity RMS over 10 to 1,000 Hz (ISO 20816-1 band), three axes	Met: IIS3DWB-class sensor, dc to 6 kHz	Datasheet and calculation; later shaker check
R6	Low vibration noise floor	0.1 mm/s RMS or better, 10 to 1,000 Hz	Met on datasheet (about 0.04 mm/s); magnet mount effect unverified	Noise calculation; later mounted test
R7	Resolve running-speed peaks	Spectrum bin 0.5 Hz or finer up to 1 kHz	Met: 0.41 Hz bins (8,192 points at 3.33 kHz)	Calculation
R8	Measure frame temperature	Within 2 °C of the surface, 0 to 100 °C	Unverified: DS18B20 is within 0.5 °C to 85 °C and about 2 °C to 125 °C (datasheet); contact error unknown	Calculation; later comparison with a thermocouple
R9	Report often enough	One summary per minute; spectrum every 15 min; delivered within 2 min	Met: about 0.4 MB per day over Wi-Fi	Data budget
R10	Install without opening live enclosures	Pod and probe in 10 min without tools; CT on an accessible insulated conductor	Not met for many machines. Pod and probe met; the CT often needs a terminal box or panel opened by a qualified person	Design review with pilot site
R11	Stay attached	No slip under 5 g peak on a painted cast frame; option for aluminium frames	Unverified: about 6 times margin estimated; steel pads included for aluminium frames	Force calculation; later pull test
R12	Tolerate hot frames	Operate on frames up to 80 °C in 40 °C ambient	Not met with the stock ABS box. Needs a stand-off or a higher-temperature enclosure; 60 °C frame is the concept limit	Thermal estimate
R13	Workshop protection	IP54 or better for the pod	Met on paper: IP54 box, IP68 glands	Design review
R14	Keep data local		Met	Design review

ID	Requirement	Target	Concept status (estimate)	Verification (TRL 3 or later)
		Works with TwinKit or any MQTT broker on the local network; no cloud account		
R15	Store data when the network is down	7 days of summaries on the pod	Met: about 10 days in 4 MB of flash	Data budget
R16	Low cost and buildable	Parts \$80 or less per machine; hand tools and a drill press; no custom PCB	Met with a thin margin: about \$79	Priced BOM
R17	Safe by design	Low voltage in the pod; certified 5 V adapter; voltage-output CT only; no connection to machine controls	Met on paper	Design review

■ Assumptions

- Target machines are driven by induction motors from 0.37 kW to 15 kW, running at 50 or 60 Hz, at speeds from about 700 to 3,600 rpm.
- A single CT on one phase is enough to judge run state and relative load on a three-phase motor whose phases are balanced; it is not an energy meter.
- The pod is mounted on a clean flat or gently curved area of the frame or bearing housing, near the non-drive or drive-end bearing, with paint thickness under about 0.2 mm.
- The workshop has Wi-Fi coverage at the machine, or a TwinKit gateway nearby.